

Empirical Workflow

Market data

STOXX 600 ESG-X
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VSTOXX; Bund 10Y
TTF; Brent; EUA

ESG spread

$$r(\text{ESG-X}) - r(\text{STOXX 600})$$

Driver transformations

ΔVSTOXX ; $\Delta \text{Bund 10Y}$
log returns: TTF, Brent, EUA

Pre-estimation checks

ADF/PP; descriptive statistics
correlations; VIF

Quantile Regression

$$\tau = 0.10, 0.25, 0.50, 0.75, 0.90$$

Moving block bootstrap

22-day blocks
robust confidence intervals

Regime analysis

2012–2019; 2020–2022
2023–2026

Rolling QR

252-day and 504-day windows

Quantile Local Projections

$$\begin{aligned}\tau &= 0.10, 0.50, 0.90 \\ h &= 1, 5, 10, 20\end{aligned}$$